

Definition. The **indefinite integral** of the vector-function $\mathbf{r}(t)$ (with respect to t) is the set of all antiderivatives of $\mathbf{r}$. If $\mathbf{R}(t)$ denotes any single antiderivative, then this set is given by

$$\int \mathbf{r}(t) dt = \mathbf{R}(t) + \mathbf{C},$$

where $\mathbf{C}$ is a *vector* of constants. Moreover, if each component is integrable on some domain $[a, b]$, then so is $\mathbf{r}(t)$ and the **definite integral** from a to b is given by

$$\int_a^b \mathbf{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle$$

Projectile Motion

A classic example of integrating a vector-function is the derivation of the equations for the motion of a projectile. Specifically, we consider an object being launched from an initial position and assume that there is no lateral movement and no extra forces at play—only gravity and the initial launch.

We assume the projectile is launched from the origin at time $t = 0$ and is launched into the first quadrant with an initial velocity of $\mathbf{v}_0$:

Notice that if $\mathbf{v}_0$ makes an angle of α with the horizontal, then $\mathbf{v}_0 = \mathbf{r}'(0) = \langle \|\mathbf{v}_0\| \cos(\alpha), \|\mathbf{v}_0\| \sin(\alpha) \rangle$. We also see that the projectile's initial position is given by the vector $\mathbf{r}_0 = \langle 0, 0 \rangle = \vec{0}$.

Now, Newton's Second Law of Motion tells us that

$$\text{Force} = (\text{mass})(\text{acceleration}) = m \left(\frac{d^2 \mathbf{r}}{dt^2} \right) = m \mathbf{r}''(t).$$

In this case, once the object is in the air, Earth's gravitational force is the only active force. This force is directed in the 'downward' direction and is given by $\langle 0, -mg \rangle$, where $g = 9.8 \text{ m/sec}^2$ is the acceleration due to gravity. Thus, we have

$$m \mathbf{r}''(t) = \langle 0, -mg \rangle = m \langle 0, -g \rangle$$

and therefore $\mathbf{r}''(t) = \langle 0, -g \rangle$. Now, we integrate!

All in all, we find that

$$\mathbf{r}(t) = \langle \|\mathbf{v}_0\| \cos(\alpha)t, \|\mathbf{v}_0\| \sin(\alpha)t - \tfrac{1}{2}gt^2 \rangle$$

This is known as the *vector equation* for the path of an ideal projectile motion. The angle α is known as the **launch angle** (or **firing angle**, **angle of elevation**) and $\|\mathbf{v}_0\|$ is the projectile's **initial speed**.

Example. A projectile is fired from the origin over horizontal ground at an initial speed of 500 meters-per-second and a launch angle of 60° . Where will the projectile be 10 seconds later?

Now, I hear you saying: Aren't projectile paths *parabolas*? The answer is yes! If you eliminate the parameter in the vector-equation, then you'll substitute $t = \frac{x}{\|\mathbf{v}_0\| \cos(\alpha)}$ into the second component of $\mathbf{r}(t)$. Doing so yields the following function:

$$y = -\left(\frac{g}{2\|\mathbf{v}_0\|^2 \cos^2(\alpha)}\right)x^2 + (\tan(\alpha))x.$$

Additionally, we can make the following conclusions:

- (i) The projectile's maximum height occurs when the vertical component of its velocity is zero. That is, when

$$\frac{d}{dt} (\|\mathbf{v}_0\| \sin(\alpha)t - \tfrac{1}{2}gt^2) = 0$$

- (ii) The total flight time occurs at the t -value for which the projectile hits the horizontal axis again. This occurs when its vertical component of its position is zero. That is, when

$$\|\mathbf{v}_0\| \sin(\alpha)t - \tfrac{1}{2}gt^2 = 0$$

- (iii) The **range**, denoted R , of the projectile is the total distance it traveled. This is found by evaluating its position vector's horizontal component at the value found in (ii):

Lastly, we note that if our initial position vector is given by $\mathbf{r}_0 = \langle x_0, y_0 \rangle$, then our projectile's path is simply shifted by these starting coordinates:

$$\mathbf{r}(t) = \langle x_0 + \|\mathbf{v}_0\| \cos(\alpha)t, y_0 + \|\mathbf{v}_0\| \sin(\alpha)t - \tfrac{1}{2}gt^2 \rangle$$